YEAR 11 MATHS METHODS INVESTIGATIVE TASK 3 SOLUTIONS

Sequences

1 **a** $t_1 = 3$ **[1]**

b $d = 7 - 3 = 4$ **[1]**

c $t_n = t_1 + (n-1)d$ **[1]**

$= 3 + (n-1)4 = 4n - 1$ **[1]**

d $t_{25} = t_1 + (n-1)d = 3 + 24 \times 4 = 99$ **[1]**

e $S_n = \frac{n}{2}(2t_1 + (n-1)d = \frac{n}{2}(2 \times 3 + (n-1) \times 4) = n(2n+1)$ **[1]**

When $n = 100$, $S_{100} = 100 \times 201 = 20\,100$ **[1]**

2 **a** 3, 6, 12, 24 **[1]**

b A geometric sequence (GP). **[1]**

c $t_1 = 3$ and $r = 2$. **[1]**

$t_n = t_1 r^{n-1}$ **[1]**

$= 3 \times 2^{n-1}$ **[1]**

d $S_n = \frac{t_1(r^n - 1)}{r - 1}$ where $n = 12$, $t_1 = 3$ and $r = 2$ **[1]**

$= \frac{3(2^{12} - 1)}{2 - 1} = 12\,285$ **[1]**

3 $S_n = \frac{n(t_1 + t_n)}{2}$ where $n = 400$, $t_1 = -5$ and $t_n = 2005$ **[2]**

$S_n = \frac{400(-5 + 2005)}{2} = 400\,000$ **[1]**

4 **a** Let $S = 1 + 3 + 5 + \cdots + 997 + 999$

$S = 999 + 997 + 995 + \cdots + 3 + 1$ **[1]**

Adding, $2S = 1000 + 1000 + 1000 + \cdots + 1000 + 1000$ **[1]**

From 1 to 999 there are as many terms as from 2 to 1000 = 2(1 to 500) = 500 terms **[1]**

$2S = 500 \times 1000 = 500\,000$

$S = 500\,000 \div 2 = 250\,000$ **[1]**

b Let $S = 1 + 3 + 5 + \cdots + (n-2) + n$

$S = n + (n-2) + (n-4) + \cdots + 3 + 1$ **[1]**

Adding, $2S = (n+1) + (n+1) + (n+1) + \cdots + (n+1) + (n+1)$ **[1]**

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From1 to n there are as many terms as from 2 to $(n + 1) = \frac{n + 1}{2}$ terms **[1]**

$2S = (n + 1) \times \frac{n + 1}{2} = \frac{(n + 1)^2}{2}$ **[1]**

$S = \frac{(n + 1)^2}{2} \div 2 = \frac{(n + 1)^2}{2}$ **[1]**

Note: n is odd so $n + 1$ is even.

5 a It is a geometric sequence (progression) with $t_1 = 1$ and $r = x$ **[2]**

b $t_n = t_1^{n-1} = 1 \times x^{n-1} = x^{n-1}$ **[2]**

c $n + 1$ (since $t_1 = x^0$, or, the nth term is x^{n-1}, the second last term.) **[1]**

d $S_n = \frac{t_1(r^n - 1)}{r - 1} \Rightarrow S_{n+1} = \frac{t_1(r^{n+1} - 1)}{r - 1}$ where $t_1 = 1$ and $r = x$

Sum of series: $S_{n+1} = \frac{x^{n+1} - 1}{x - 1}$ **[2]**

e There are 20 terms ($n = 19$).

$S_{20} = \frac{x^{20} - 1}{x - 1}$ **[1]**

f $1 + x + x^2 + x^3 + x^4 + \cdots + x^{19} = \frac{x^{20} - 1}{x - 1}$

Multiply both sides by $(x - 1)$.

$x^{20} - 1 = (x - 1)(1 + x + x^2 + x^3 + x^4 + \cdots + x^{19})$ **[1]**

g If you had n instead of 20, $x^n - 1 = (x - 1)(1 + x + x^2 + x^3 + \cdots + x^{n-1})$. **[1]**

Proof:

$S_n = 1 + x + x^2 + x^3 + x^4 + \cdots + x^{n-1}$

$xS_n = x + x^2 + x^3 + x^4 + \cdots + x^{n-1} + x^n$ **[1]**

Subtract the top line from the bottom line.

$(x - 1)S_n = x^n - 1$ **[1]**

$S_n = 1 + x + x^2 + x^3 + \cdots + x^{n-1}$

$\therefore (x - 1)(1 + x + x^2 + x^3 + \cdots + x^{n-1}) = x^n - 1$, or,

$x^n - 1 = (x - 1)(1 + x + x^2 + x^3 + \cdots + x^{n-1})$. **[1]**

6 a Growth is a geometric sequence. $t_1 = 20$ and common ratio $r = \frac{16}{20} = \frac{4}{5}$ or 0.8.

[1 mark for GP and 1 mark each for t_1 and r]

b i Continue the pattern for growth. …10.24, 8.192, 6.5536

Answer: 6.5536 cm.

OR

$t_1 = 20$ and r $= 0.8$ so $t_n = 20 \times 0.8^{n-1}$.

$n = 6 \Rightarrow t_6 = 20 \times 0.8^5 = 6.5536$ cm. **[1]**

ii $t_1 = 20$ and r $= 0.8$ so $t_n = 20 \times 0.8^n$. **[1]**

c i Either add the growths to the initial height (0.7 m = 70 cm).

Answer: $70 + 20 + 16 + 12.8 + 10.24 + 8.192 + 6.5536 = 6.5536$ cm.

OR

$t_1 = 20$ and $r = 0.8$ so $S_n = 70 + 20 \times \dfrac{1 - 0.8^n}{1 - 0.8} = 70 + 100(1 - 0.8^n)$.

$n = 6 \Rightarrow t_6 = 70 + 100(1 - 0.8^6) \approx 143.79$ cm or 1.44 m.

ii $t_1 = 20$ and r $= 0.8$ so $S_n = 70 + 20 \times \dfrac{1 - 0.8^n}{1 - 0.8} = 70 + 100(1 - 0.8^n)$.

[1 mark for each part]

d Note that as $1 - 0.8^n < 1$ then height $< 70 + 100 \times 1 = 170$ cm or 1.7 m.

OR (Not on course but some students may work it out)

Use growth that is a GP with $t_1 = 20$, $r = 0.8$.

Take the limit as $n \longrightarrow \infty$.

$$S\infty = \frac{20}{1 - 0.8} = 100 \text{ cm} = 1 \text{ m}$$

The bush was already 0.7 m so total height will be 1.7 m.

The light will not be blocked out by the bush. **[2]**

7 a Let there be n posts. There will be n sections of 25 cm (post + space) and an additional 5 cm for a post at the end.

$25 \times n + 5 = 505 \Rightarrow n = 20$ so 21 posts including the one at the start. **[2]**

b The posts are an AP. As the ground is sloping uniformly the posts will need to increase uniformly in length, i.e. the difference in length between any two posts will be the same (constant), which is the definition of an AP. **[2]**

c $40 + 15 = 55$ cm **[1]**

d $50 + 15 = 65$ cm **[1]**

e $t_1 = 55$, $t_n = 65$ and $n = 21$ **[1]**

$t_n = t_1 + (n - 1)d$

$65 = 55 + 20d$ **[1]**

$d = 0.5$ **[1]**

Each post will be 0.475 cm longer than the previous post. **[1]**

f Vertical: An AP with $t_1 = 55$, $d = 0.5$ and $n = 21$

$$S_n = \frac{n}{2}(2t_1 + (n - 1)d) = \frac{21}{2}(2 \times 55 + 20 \times 0.5)$$

$= 1260$ cm or 12.6 m **[2]**

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Horizontal: $2 \times 505 = 1010$ cm $= 10.1$ m **[1]**

Total length of wood needed $= 1260$ cm $+ 1010$ cm

$= 22\,270$ cm $= 222.7$ m **[1]**

g Use 2×5.1 m lengths for the horizontal lengths. The two spare 5 cm pieces cannot be used.

The common difference of the vertical pieces d is 0.5 cm.

Thus our goal is to waste the minimum total length in cutting the vertical posts from the lengths available.

Students can use a spreadsheet and trial and error.

Solution 1

Starting with the shortest pole, $S_n = \frac{n}{2}(2a + (n-1)d)$ where $a = 55$ and $d = 0.5$.
You could use the spreadsheet below.
Enter the formula for the sum of an AP in cell D2: =0.5*C2*(2*A2+(C2-1)*B2)

	A	B	C	D
1	a	d	n	total
2	55	0.5	7	395.5

Trying different numbers, the least wastage is for 5 posts giving 280 cm, wasting 20 cm from a 3 m length.
This leaves 16 posts.
There are 16 pieces left, starting at the 6th piece.
The length of the 6th piece is $a + (n-1)d = 55 + 5 \times 0.5 = 57.5$ cm
Following the same process the next 8 pieces can be cut from a 4.8 m length, wasting 6 cm.
The last 8 can be cut from a 5.1 m length, wasting only 4 cm.
Altogether Ben needs 3 at 5.1 m, 1 at 4.8 m and 1 at 3 m, wasting a total of 40 cm.
Wood used is $3 \times 5\,\text{m} + 2 \times 4\,\text{m} = 23\,\text{m}$
Other solutions are possible that waste small amounts of timber.

[10 marks]

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